

the potential implementation issues.

For complete and successful implementation of the smart grid, India needs a national vision and a flexible plan. The country needs to develop policies and regulations to create a receptive environment for smart grids by encouraging innovation, establishing standards for interoperability and allowing market-oriented solutions.

How is IEEE positioned to lead the smart grid initiative? What does it do in smart grid standards?

Smart grid is a strategic area of focus for IEEE-SA globally, and IEEE-SA has more than 100 standards and standards in development relevant to the smart grid. India ranks as the third largest market for smart grid investments and there is initial momentum in this area with the formation of the Smart Grid Task Force. What India lacks is a collaborative environment that works towards setting global standards and regulations so that the smart grid can use interoperable technologies and become a reality faster. It is this gap that IEEE-SA aims to address as one of the world's leading and most credible standards setting bodies.

It is critical that India invests in collaboration and creation of standards at this initial stage. A fragmented approach will result in loss of economies of scale and delay in the emergence of the smart grid industry.

How is IEEE-SA working to help with smart grids in India?

Industry standards are among the most important foundational elements and form the basis of a plug-and-work architecture. The smart grid needs standards and practices that integrate intelligent equipment across not just a network but multiple diverse industries.

IEEE-SA is actively growing its engagement in India to educate, promote and help evangelise smart grid standards interests in the country. We have begun the initial steps towards engaging India and its professional technical community with initiatives such as formation of a Standards Interest Group (SIG) for India that will provide a platform for the Indian technical community in global standards development including those for the smart grid. Outreach programmes like smart grid workshops deliberating the role of standards and challenges in the Indian context are also being conducted across cities.

IEEE-SA recently published IEEE 1901 broadband-over-power-line (BPL) communications standard that enables communication data rates in excess of 500 Mbps. For the fast growing Indian market, BPL technology will enable access to affordable communication to the country's vast rural network. Despite India's power grid being not as advanced as developed countries', it is likely to make an impact as its benefits far outweigh its drawbacks. Corporate India could quickly embrace BPL because of its inherent advantages over Ethernet. At present, broadband connection has a speed of 100 Mbps, but BPL promises 200 Mbps and an easy-to-use

connection within 30 minutes.

Apart from consumer-based advantages, other opportunities include automatic energy meter reading, real-time system monitoring, preventive maintenance, voltage control, outage detection and restoration, load management on the power grid, load scheduling, load forecasting, capacitor bank control and development of smart grids, which could add to conservation of energy and improve system reliability, service and safety for electricity customers.

IEEE-SA will also publish the IEEE P2030 standard and the IEEE guide for smart grid interoperability of energy technology and information technology operation with the electric power system, and end-use applications and loads later this year.

What kinds of problems are you facing here?

Too many. The significant ones are low metering efficiency, adding power capacity in support of a projected energy growth, need to interconnect regional grids, need to build an intelligent grid, minimising transmission and distribution losses, poorly planned distribution networks, overloading of system components, lack of reactive power support, power theft and inadequate grid infrastructure.

What are your key programmes under the IEEE smart grid initiative?

IEEE-SA is leveraging its strong technical foundation to develop Smart Grid standards, share best practices, publish developments and provide related educational offerings to advance technology and facilitate successful smart grid deployments throughout the world.

How these help organisations with their R&D activities?

They help them adopt accelerated smart grid technologies, minimise technology obsolescence risks and access the capital required for this transition at reasonable cost.

If everything goes well, what timeframe do you see for effective deployment of smart grid in India?

In my opinion, it will take India at least five years to realise the full impact of smart grid when a utility control room operator can regulate an electric meter in your home. Again, it will depend on the cooperation among the various stakeholders (regulators, utilities, vendors, customers, etc). Everybody has to work together and move at the same speed.

The technology can help us reduce the electricity transmission and distribution losses to 5-10 per cent annually. But there is much more involved in adopting the smart grid; for instance, in cities like Mumbai, there is a serious problem of theft of electric power, which needs to be addressed. Regulatory controls are needed. Without smart grid, India will not be able to keep pace with the growing needs of its cornerstone industries and will fail to create an environment for growth of its highly technological and telecommunications sectors. ●